

DE.AUTO

CHAPTER 5

THE TRANSMISSION

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English



POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE
MUST-KNOW TERMS

Car modifications, explained

Chapter Five: The Transmission

“COUNT THE GEARS. FEEL THE SHIFT.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

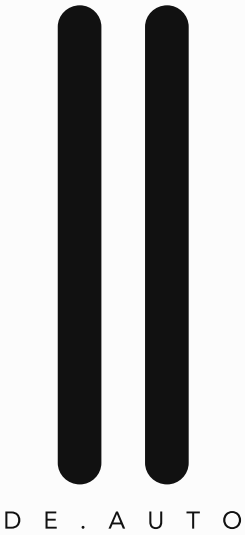
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

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READY?

Your neighbour says: “Mine’s a 10-speed!” And your friend says: “CVT? I hate it!” And you just stay quiet and nod? 😅

By the end of this chapter you'll know:

- ✓ What does the gearbox do, and why does a car need one at all?
- ✓ What does “gears” mean, and why do some cars have 4 gears and others have 10?
- ✓ The difference between the manual gearbox, the automatic, the CVT and the DCT.
- ✓ Dry and wet clutches, and the signs that a clutch is wearing out.
- ✓ The real difference between straight and helical gears, meaning Straight-Cut versus Helical.
- ✓ Common gearbox mistakes, and the truth behind the myth of shifting into neutral (N).



01 What is a gearbox? 🕹️

The gearbox (Transmission) is the middle-man between the engine and the wheels. It takes the power from the engine and changes its **ratio** to suit your speed and conditions: want to launch? It gives you a push. Want to fly down the highway? The engine relaxes and the car moves fast.

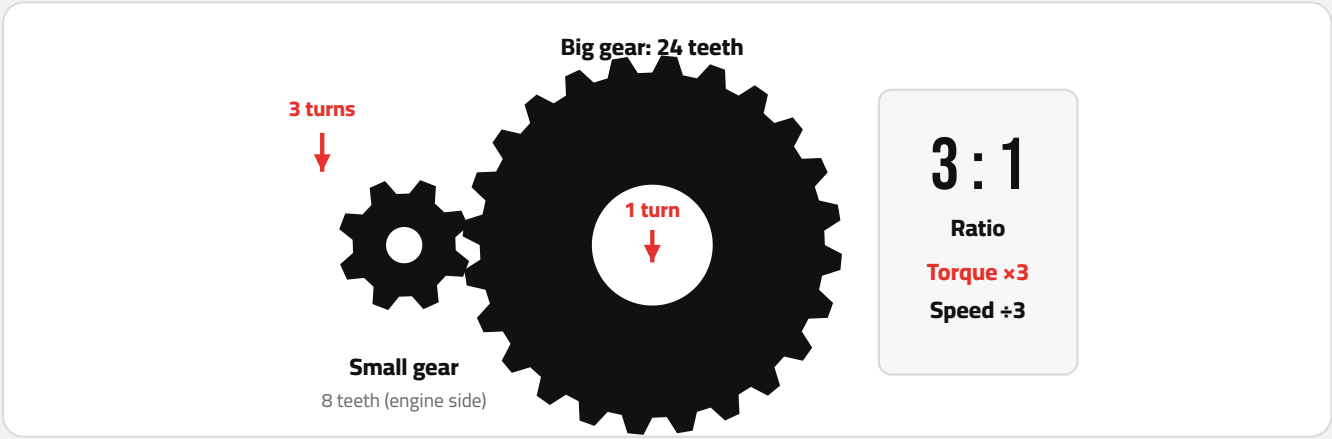
Why does a car need a gearbox at all?

The engine works best in a certain rev range, and it can't turn from zero and give strong power. **And the car starts from standstill.**

Launching needs a lot of force, and high speed needs a calmer engine. Without a gearbox, you get either a weak launch or an engine screaming on the highway.

The engine has to keep running even when the car is stopped, so we need something to disconnect it from the wheels (a clutch or a torque converter).

The secret of the game: a small gear drives a big gear ⚙️



An illustration: a small gear turns 3 times so that the big gear turns once

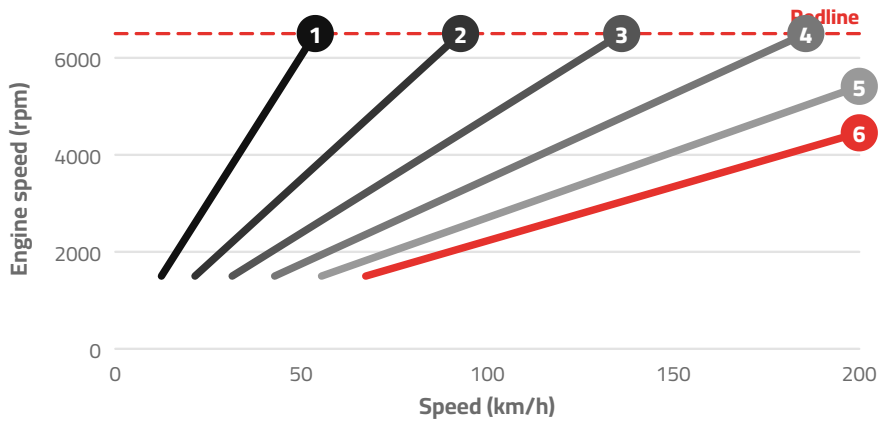
3 : 1

It means: the engine turns 3 times while the wheel turns once. You push with roughly ×3 more force (ignoring losses), but at a lower speed. Swap the two gears and the picture flips: more speed and less force. 🔁

> 🔢 **The rule:** The ratio = the number of teeth on the driven gear ÷ the number of teeth on the gear driving it. That is all the "science" of gears, and the rest is detail.

02 Gears: tall and short 📏

Let's sort out the word that confuses everyone. **First gear is short** (its ratio is numerically high, like 3.8 to 1): big pushing force and low speed. **The last gear is tall** (its ratio is small, like 0.7 to 1): the engine spins calmer and you go faster at the same revs.



An illustration with assumed numbers (six gears, a 3.9 final drive, a 65 cm tire diameter). The numbers differ from car to car

❌ The common saying "the faster you go, the taller the gears you need" is confusing. More precisely: as your speed goes up, you **move up to a gear with a higher number**, and that gear is **taller** (its ratio is smaller). So sixth gear is taller than first, not shorter.

14,500 revolutions per minute, roughly. That is what the engine would have turned at if you tried to do 120 km/h in first gear in this example. No engine could take that! That is why we need gears. 😮

So why do some gearboxes have 4 gears and some have 10?

More gears = smaller steps

Each shift changes the revs only a little, so the engine stays close to its ideal range, and acceleration comes out smoother.

And taller gears for the highway

A very tall gear lets the engine spin calmly on the highway, which cuts fuel consumption and noise.

Older gearboxes

3 to 5 gears were enough for the engines of that time, and they were simpler and cheaper.

The price

More gears mean more weight, complexity and cost, and harder programming to choose the right gear.

03 The four types of gearbox 🧩

They all do the same job, but each has its own method and personality:

Manual gearbox
Your left foot is on the clutch and your hand on the lever. Full control and a direct feel.

Automatic (Torque Converter)
A torque converter with oil, plus planetary gears. It picks the gear by itself, and it is the most widespread.

Continuously variable (CVT)
Belts or a chain and two cone-shaped pulleys. There are no fixed gears, and the ratio changes smoothly.

Dual-clutch (DCT)
Two clutches, one for the odd gears and one for the even. It gets the next gear ready before the shift.

Quick comparison

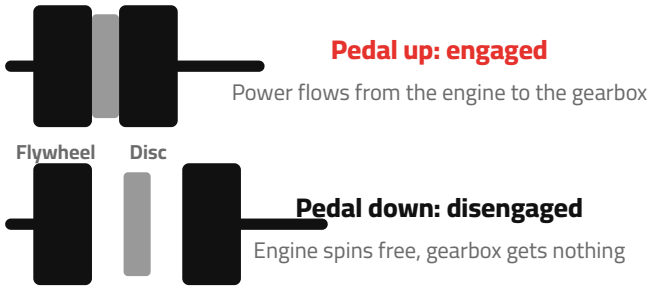
Type	The idea	The good	The bad
Manual (MT)	A clutch you press yourself	Fun and control, and usually lighter and simpler	Tiring in traffic and takes skill
Automatic (AT)	Torque converter + gears	Comfort, smoothness and lots of options	Sometimes heavier, with less feel
CVT	Two pulleys and a belt	Smoothness and fuel saving	A “rubber-band” feel for some people
DCT	Two clutches	Very fast shifting	Sensitive to heavy stop-and-go traffic

💡 **Watch the word:** In everyday talk, the word “automatic” covers the last three, because the driver doesn’t press a clutch. But a gearbox engineer means by AT specifically the type with a torque converter.

🎧 **The deal with the CVT:** Why do some people hate the CVT and others love it? The reason is mostly the **feel**: the engine raises its revs first and then the speed catches up, which gives a sound like “stretching a rubber band”. Modern designs have improved this a lot, and your experience depends on the model.

04 The manual gearbox and the clutch 🦶

In a manual gearbox, **the clutch** is the link between the engine and the gearbox. Press the pedal and they disconnect so you can change gear; release it and the power reconnects.



A simplified clutch drawing: the flywheel (left), the friction disc and the pressure plate

Dry clutch and wet clutch 💧

Dry (Clutch)
A disc that works by friction in air. Simpler and lighter, and it wastes less energy. It is the usual choice in most manual gearboxes.

Wet (Clutch)
Multiple discs bathing in oil that cools them. It handles more heat and torque, and you find it in automatic gearboxes, some dual-clutch units and motorbikes.

Signs the clutch is getting tired ⚠️

Slipping: the engine revs rise and the car doesn't accelerate as hard (it feels like it is "slipping away").

A burning smell like burnt paper, especially after climbing a hill or sitting in traffic.

The bite point has moved up: the clutch grabs at the very end of the pedal travel instead of the middle.

Vibration or difficulty getting into gear plus a grinding sound, and sometimes the pedal sticks or returns slowly.

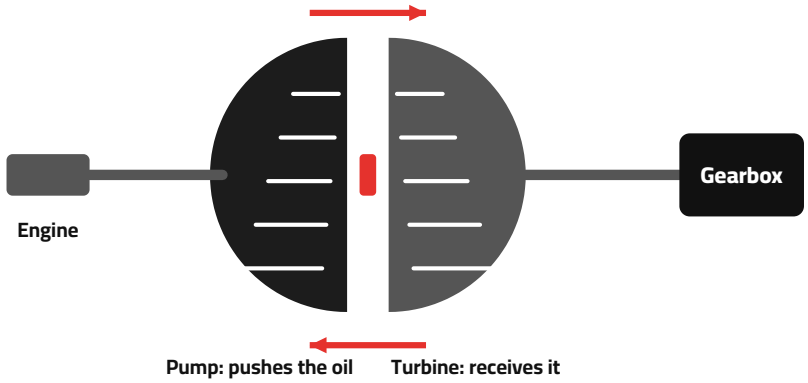
✅ **DE.AUTO tip:** Don't rest your foot on the clutch pedal, and don't hold the car on a hill with "half clutch"; use the brake or the handbrake. Repeated half-clutch use heats the disc and shortens its life.

✕ **MYTH**
If you can't learn the clutch, you're a poor driver.

✓ **FACT**
The clutch is a skill you learn by repetition, and most people get it within a few days of practice.

05 The automatic: two fans and some oil 🌀

Imagine two fans facing each other: one is powered by electricity, so it blows air that spins the second fan without any contact. That is the idea of the **torque converter**, but with oil instead of air.



A simplified idea: the pump (left) pushes the oil, the turbine (right) spins, and the red centre is the stator that redirects the oil

Pump and turbine

The pump is tied to the engine and the turbine to the gearbox. The oil between them is the bridge, which is why the car can stop with the engine running and no clutch.

The stator

It redirects the returning oil to give an extra push at launch, which is where the name "torque converter" comes from.

The lock-up (Lock-up Clutch)

At steady speeds the converter locks mechanically, so slip losses drop and consumption improves.

Planetary gears

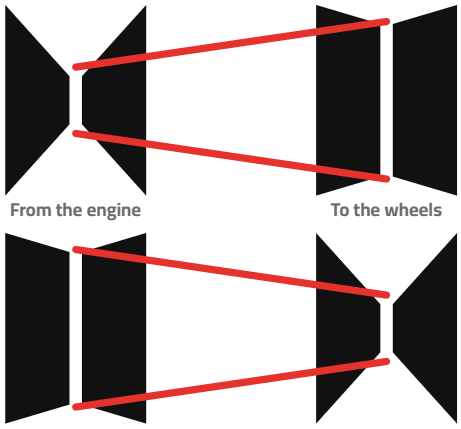
Gear sets whose ratio changes by holding parts with internal clutches, and they are what give you the 6, 8 and 10 gears.

The automatic letters you must know

Letter	Name	Meaning
P	Park	Locks the gearbox with an internal pin; use it after a complete stop
R	Reverse	Backwards
N	Neutral	No connection between the engine and the wheels
D	Drive	Shifts by itself, going forward
M / +/-	Manual mode	You choose the gear and it protects you

06 The CVT: a gearbox with no fixed gears

The CVT (Continuously Variable Transmission) has no fixed gears to shift. Instead it has **two cone-shaped pulleys** and a belt or chain between them. The pulley widens and narrows, so the belt rides at a different radius, and the ratio changes continuously, with no “shift”.



Launch
Small engine pulley

Cruising
Large engine pulley

A simplified drawing: the top row is moving off from standstill and the bottom is road speed. The red belt moves between the small and the large radius

∞ roughly the number of possible ratios between the shortest and the tallest ratio, because it moves smoothly without steps. That is the secret of its smoothness and economy.

Why does it save fuel?
It can keep the engine at revs that suit low consumption, and it reduces the jolts between gears.

Why is it criticised?
The revs rising before the speed catches up gives a different feel, and its sound can seem tiring to someone used to a regular automatic.

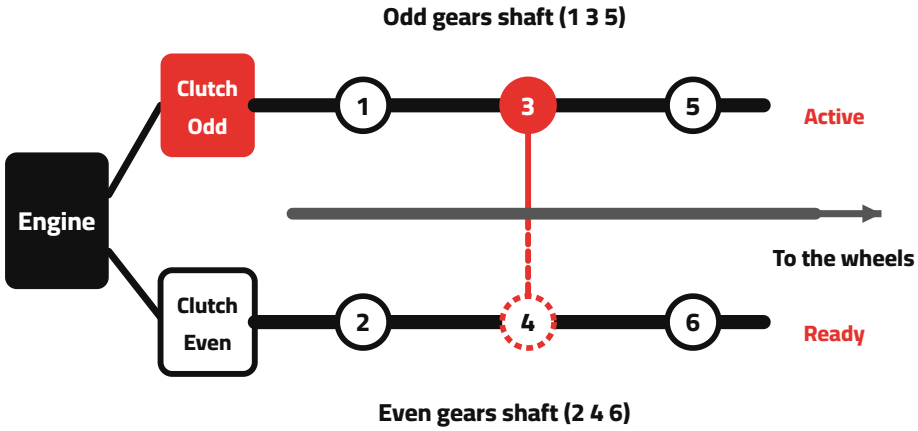
Simulated gears (Simulated Steps)
Many manufacturers program the CVT to imitate fake “shifts” so the driver gets the familiar feel.

Maintenance
It usually needs a special oil from the manufacturer. Follow the service schedule in your owner’s manual to the letter.

💡 **A fair warning:** The durability reputation of some CVT models varies from one manufacturer and generation to another. Read owners’ experiences of the model you’re considering before you buy, instead of judging the whole type.

07 The DCT: two clutches and speed 🏁

The dual-clutch gearbox (Dual-Clutch Transmission) is like a manual gearbox, but the computer works the clutch, and instead of one clutch it has **two**: one holds the odd gears (1, 3 and 5) and the other the even gears (2, 4 and 6). And here is the trick 👇



A simplified drawing: the car is running in 3rd gear and the computer has 4th ready on the second shaft

While you drive in gear **3**, the computer already has gear **4** ready on the other shaft. At shift time it releases one clutch and engages the other almost at the same instant, so the shift is very fast with no break in the power to the wheels.

Dry (DCT)

Usually in lighter gearboxes with less torque. Efficient and light, but heat builds up during slow slipping.

Wet (DCT)

Discs in oil that cools them. Usually in stronger, higher-torque gearboxes, and they handle heat better.

Why do they say it is sensitive to traffic? 🚦

In heavy traffic and slow movement (creep and stop), the computer makes the clutch **slip partially** so the car moves smoothly, like you do with half clutch. That repeated slipping generates heat and wear, especially in the dry type. So if your whole day is traffic, think about the type before buying.

✅ **DE.AUTO tip:** Stick to your owner's manual: many manufacturers advise holding the car on the brake (not the throttle) on hills, and using neutral or park during long stops. It depends on your model.

08 Which is faster: manual or automatic? ⚡

✕ MYTH

The manual gearbox is always faster, and the automatic is for lazy people.

✓ FACT

This used to be true. Modern gearboxes (automatics with many gears, and the dual-clutch DCT especially) often match a manual in acceleration or beat it, because they shift faster and more precisely than any human. The manual wins on **fun and control**, not necessarily on the numbers.

Buttons behind the steering wheel: shifters and paddles 🎛️

Many automatics and DCT transmissions have a **manual mode**: you shift yourself with the lever or with buttons behind the steering wheel called Paddle Shifters. The feel of a manual, without the clutch.

What do you gain?

You choose the gear before a corner or a hill, and you control engine braking.

And am I free to do anything?

Mostly no. The computer protects you: it refuses a gear that would push the engine past its limit, and shifts for you if you reach the limit.

No clutch pedal

In automatics and the DCT there is no pedal at all, and the computer operates the clutch. And some race cars use special sequential gearboxes that shift with a pull.

"Launch control"

The idea: the computer sets up the launch from standstill in the best possible way. This is only the concept; we explain it in Chapter 13, and it belongs on dedicated tracks only.

> 💡 Some automatic cars drop a gear by themselves as you brake before a corner, so the engine is ready to accelerate after it.

🎯 Challenge 1 **While cruising at a steady speed, which gear gives lower engine revs?**

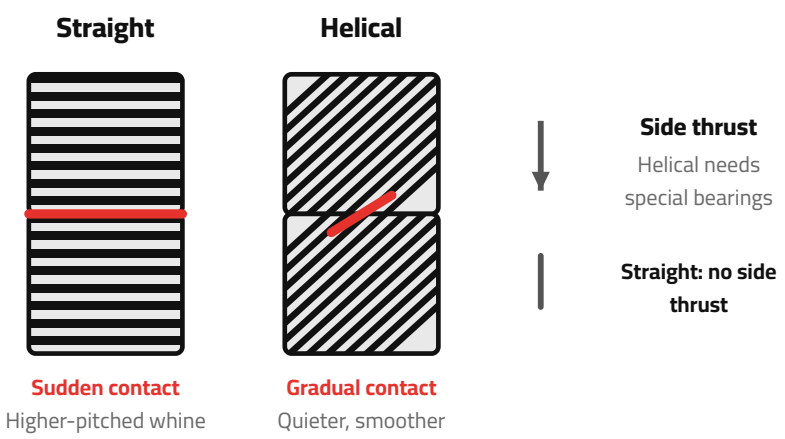
A) First gear | B) The highest-numbered gear (the tallest) | C) Neutral

🎯 Challenge 2 **In the dual-clutch gearbox, the next gear is...**

A) Waiting for the shift to finish | B) Ready on the other shaft | C) Missing

09 Straight gears and helical gears 🔧

The proper names for the comparison: **straight gears** (Straight-Cut / Spur) versus **helical gears**. The difference is the tooth shape: a straight tooth is parallel to the axis, and a helical tooth is angled.



A side view of two meshing gears. The red line is where the teeth make contact

	Straight (spur)	Helical
Sound	Loud “whine” when moving	Quieter and smoother
How they make contact	The whole tooth makes contact at once	Starts at one end and creeps along, so more teeth are engaged at the same time
Strength	Strong and durable, but its size is limited by its load	Usually handles a higher load at the same size
Friction	Lower, and slightly higher efficiency	More sliding friction, and slightly more energy loss
Side thrust	None	Yes, and bearings that can take it are needed

💡 Why do race gearboxes (like the Dog-Box) often use straight gears? Because they are simpler with less friction, they don’t create side thrust, and they suit dog engagement that shifts fast without synchronisers. The price: loud noise and a roughness that doesn’t suit daily use.

💡 **Did you know?** The reverse gear in many manual gearboxes is straight-cut, which is why you hear a light whine when you drive backwards.

10 Myths and common mistakes 🚫

✕ MYTH	✓ FACT
"Stopping the car suddenly while in N ruins the gearbox".	The brakes are what stop you, and a gearbox in neutral carries no load. This claim isn't accurate. The real danger of neutral is that you lose **control and engine braking**, and your reaction is delayed.
✕ MYTH	✓ FACT
"I put the car in N while driving to save petrol".	In most modern cars, when you lift your foot and the car is in gear, the engine cuts the fuel almost completely. In neutral it goes back to burning fuel to idle. So you don't save, and you lose control. Some modern cars have a "coasting" mode that disconnects the engine by itself, but that is the computer's designed decision, not yours.

Mistakes not to make ❌

❌ **Selecting Reverse (R) or Park (P) while the car is still moving.** In an automatic it can cause a jolt and wear on the gearbox or the park pin, and in a manual it is hard to engage reverse and it grinds. Safest: come to a complete stop, then change. (Limits vary and modern systems block some of these, but don't rely on them.)

❌ **Driving with a "dying" clutch or getting used to half clutch.** Slipping burns the disc, and the longer you wait, the more the repair costs (usually the whole kit is replaced).

❌ **Holding the car on a hill with the throttle in an automatic.** The computer lets the converter slip and heat builds up. Use the brake or the handbrake.

❌ **Revving in neutral and then slamming into D suddenly.** It gives the gearbox and driveshafts a harsh shock, and they wear with repetition.

✅ **A simple rule: Come to a complete stop, then change mode.** A few seconds protect you from an expensive repair.

11 Gearbox oil and its maintenance 🛢️

The gearbox lives on oil: it lubricates the gears, cools them, and in automatics, CVT and wet DCT units it also works as the hydraulic medium. Each type has its own specific oil.

Manual gearbox
Gear oil with a viscosity set by the manufacturer.

Automatic
Uses ATF fluid to a defined specification.

Continuously variable
Needs a dedicated CVT fluid, and no other will do.

Dual-clutch
A special oil for wet gearboxes, and gear oil for dry ones.

The wrong oil makes shifting rough or makes it slip, and damages the gearbox. Stick to the specification written in the manual.

Some manufacturers call it “lifetime”, and many technicians advise changing it periodically. Ask a trusted workshop what suits your car and the way you use it.

Signs that need a check: red or brown oil under the car, a burning smell, delayed shifting, vibration, or a strange whining sound.

Heat is the gearbox’s enemy. Traffic, heavy towing and hills heat it up, so care matters more if your use is harsh.

Garage talk vs the proper names 🗣️

Gulf garage talk	Proper name	What it means
“Numbers” / speeds	Gear ratios	Each ratio between the engine and the wheels
Gear / “jeer”	Transmission	The box that holds the gears
Debriyaj	Clutch	Disconnects and reconnects the engine and the gearbox
The kit	Clutch kit	Disc + pressure plate + release bearing
“Cross-cut” / “street-cut”	Helical / straight-cut gears	“Cross-cut” is not a correct term for gears

> 💡 Workshop names vary from region to region, so if you hear a name you don’t understand, ask the technician to show you which part he means.

12 Which gearbox suits you? 🎯

What you want	Closest for you	Note
Want fun and control?	Manual	Gives you the purest feel of driving
Want comfort and traffic?	Automatic	Best suited to everyday use for most people
Want smoothness and economy?	CVT	Try it before you decide; the feel is personal
Want speed and response?	DCT	Excellent on open roads, watch out for heavy traffic

FIRST fully automatic gearbox in mass production appeared many decades ago, and it used a fluid coupling. 🏭

💡 **Did you know?** Modern computer-linked gearboxes learn your driving style and adjust shift timing. And some race cars have sequential gearboxes that shift with a push, with no clutch pedal.

🎯 Challenge 3 **Which statement is more accurate about a "tall gear"?**

A) Its ratio is large and its force is high

B) Its ratio is small and its revs are lower at the same speed

C) It isn't used on the road

🎯 Challenge 4 **Which of the following is true about the helical gear?**

A) Quieter and smoother, but it produces side thrust

B) Always louder

C) Never used in cars

🎯 Challenge 5 **Why might the DCT (especially the dry type) struggle in heavy traffic?**

A) Because it has no oil

B) Repeated partial clutch slipping generates heat

C) Because it has 3 clutches

➡️ SOON The gearbox transmits the power... but to which wheels? Front? Rear? All four? And why do some cars spin one wheel and leave the other digging in? What is the role of the **locked differential** and the LSD? All of that is in **Chapter 6: Drive Systems and Torque**. 🛞🔥

Wrap-up 📌

Now you know the gearbox is the smart middle-man between the engine and the wheels, and that “gears” are just ratios between toothed wheels: short for launching and tall for the road. You’ve seen how the manual works with its clutch, the automatic with its oil converter, the CVT with its two pulleys, and the DCT with two clutches. And you can tell straight gears from helical ones. 🚀

🎯 Challenge answers:

1) The highest-numbered gear (the tallest).

2) Ready on the other shaft.

3) Its ratio is small and its revs are lower at the same speed.

4) Quieter and smoother, but it produces side thrust.

5) Repeated partial clutch slipping generates heat.

➡️ In **Chapter 6: Drive Systems and Torque**, we see where the power goes after the gearbox: FWD, RWD and AWD, and the secrets of the differential. Ready?